

ABSTRACT

The SRIHER is a Deemed University, where Multiple Facultys are there. In this a Phd student has approached me to Gamify the education and teaching methods for their students. So for our Own product and apart from Traditional learning, a quiz webapp is developed be me. The developmentof Quiz web app undergone several changes in between and apart from that, most of the quiz apps dosen't focus on general education and customizable options. But our app comes in and shows it Individual Discipline's unit based learning, where Student of a particular discipline, for example, Nursing can attend quizzes from there curriculum as units. This will be a unique feature for this web app. Also a multi disciplinary app fro mobile is under development, and this web app will be provided to the Phd Student for their Data Collection. So this will be SRIHER's own product and it will be puvlished, so that all students and public can access for.

CHAPTER 1

INTRODUCTION

1.1. Need for the Study/Technology

This Study is totally for analyzing the difference between the Traditional Learning and Gamified Learning. This Web application ensures that there will be interactivity between the students and subjects. Multiple papers were analyzed and the need for this study is, even though multiple systems and techniques exist, nowhere a proper enterprised system is established for fully digital education which is interactive. So this study aims for the Good results of the students using this quiz app.

1.2. Applications

Applications are,

- Interactive Learning
- Experience points-based level progression
- User centric customization
- Day to Day Leaderboard Update
- Different Question Modes
- Domain based Units
- Live Quiz
- And more...

1.3. Motivation behind the Study

The Motivation is that a study on how the Gamified or interactive learning implies effective learning among students. As an existing Technology, Kahoot, way ground and Mentimeter gives various features and interactivity

for students. This makes the learning heartfelt and easy to remember. So, this motivates to create a Unit based quizzes and level-based progression to consistent incoming of specific Department students irrespective of their syllabus. Thus, Motivation is to study the effectiveness

1.4. Challenges to be addressed

- Handling Multiple Users at once (Scaling)
- Creating Creative Question Modes
- Selecting Suitable UI
- Thinking of creative features to add
- Making it portable for Multi Device
- Designing User-centric customizations
- Implementing security measures for security of the website

CHAPTER 2

2.1 LITERATURE SURVEY - Minimum 15 Works

CHAPTER 3

PROBLEM STATEMENT AND OBJECTIVES

3.1 Problem Statement

Traditional Nursing teaching techniques depends on the traditional learning methods that is, lectures, pdfs, tests using papers, and that earns low student engagement, lack of real time competition, no feedback system, especially in skill required scenarios

→Poor in engagement

→No competition

→Lack of Questionnaire types

→Instable accessibility

To solve this problem,

- A websapp or an website with Nursing specific learning units and geneal quiz builder is introduced and any nursing students can attend the units irrespective of their curriculum.
- Tech Stack will be, React+vite and the middleware will be Node.js+Express.js. REST API handles the routes of api, middleware and etc and using DB as Supabase.
- Web-sockets are used to enable live multiplayer quiz sessions, that teachers hosts and students join and play

3.2 Objectives

Objective 1:

To design a Full stack Web application which is a quiz platform that is specified for Nursing education with interactive gamified elements with React, Node, Express and a remote DB with multi-role access control, attractive dashboards and a understandable quiz management system.

Objective 2:

To establish a Multi questionnaire mode Quiz builder, which has, MCQs, Video based, audio based, jumbled letter, sequence ordering, matching, etc., giving variety of experience and engagement to the students rather than the text only quizzes

Objective 3:

To combine Domain specific elements like clinical quizzes and unit based learning, scenario based quizzes and relevant gamification frameworks like XPs, leaderboards, streaks and badges

Objective 4:

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness

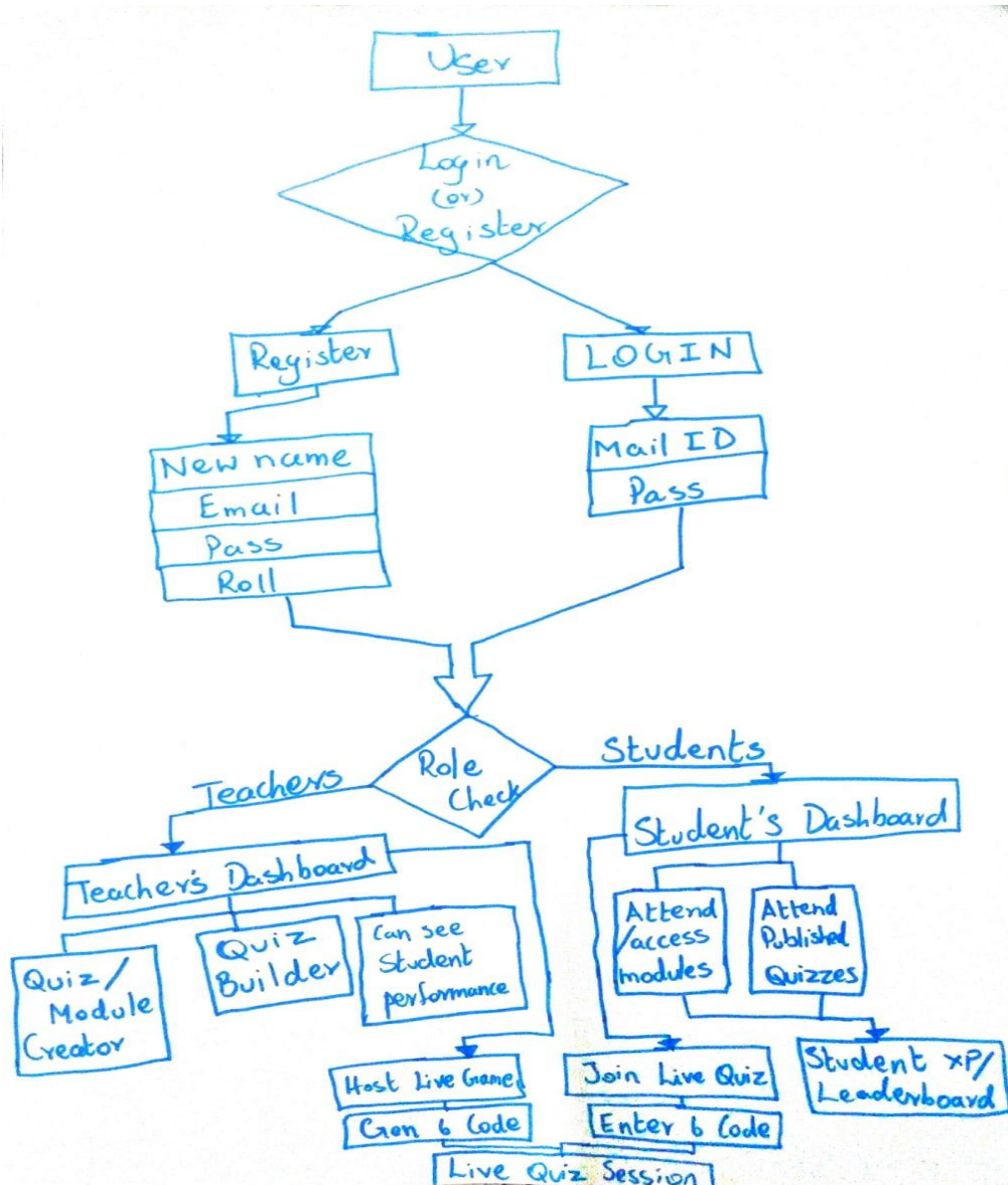
Objective 5:

At last, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

CHAPTER 4

METHODOLOGY

4.1 Workflow Diagram



4.1.1 – Workflow Diagram of the Webapp with Role based control

This Diagram shows the Flow of the Website.

The user logs in or signups through the Login page and the Role of the user is checked and they are landed to the respective dashboards,

Teachers have the ability to draft and assign Quizzes and and also have the ability to host live quizzes.

Also they can view the Leaderboards and all.

In the other side, Student's Dashboards have their personal streaks and completion history and recently attended quiz. They can play a mini-game (Which is not ready) and can view leaderboard and attend Units through Unit section. They can join a live game using the code that is displayed.

4.2 Modules

Frontend Module (UI layer):

Method/Algo: SPA is based on the Component and routing is done on the client side and this manages context API state

Tech (S/W AND H/W): React 19, vite, React Router, Three.js, Lottie

Implementation: 11 pages, again usable components, 2 context providers, API Client

Backend Module (Server):

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

Auth Module (Security):

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

DB Module (Data):

Method/Algo: RESTful API, and MVC Pattern with middleware chaining. Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORS

Implementation: 5 route modules, endpoint for file upload, media handler

Real-Time Module (Security):

Method/Algo: RESTful API, and MVC Pattern with middleware chaining.

Uploads go through MIME Filtration

Tech (S/W and H/W): Node.js, Express.js, Multer, CORs

Implementation: 5 route modules, endpoint for file upload, media handler

4.2.1. Objective -1 (Repeat similarly to all 4 or 5 objectives)

To design a Full stack Web application which is a quiz platform that is specified for Nursing education with interactive gamified elements with React, Node, Express and a remote DB with multi-role access control, attractive dashboards and a understandable quiz management system.

4.2.2. Objective 2:

To establish a Multi questionnaire mode Quiz builder, which has, MCQs, Video based, audio based, jumbled letter, sequence ordering, matching, etc.,, giving variety of experience and engagement to the students rather than the text only quizzes

4.2.3. Objective 3:

To combine Domain specific elements like clinical quizzes and unit based learning, scenario based quizzes and relevant gamification frameworks like XPs, leaderboards, streaks and badges

4.2.4 Objective 4:

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness

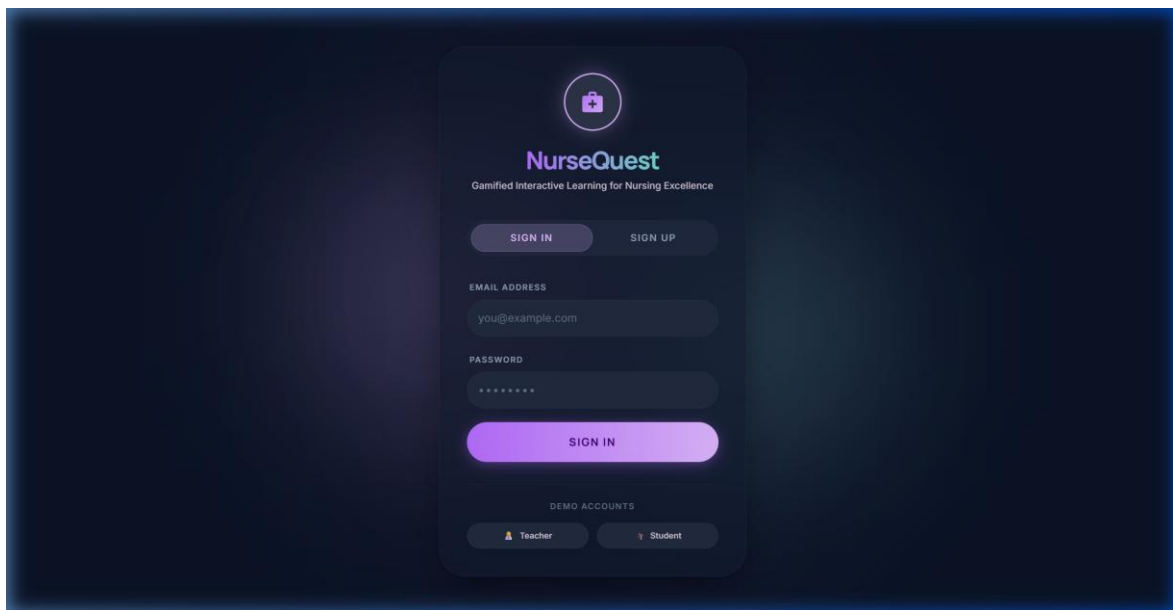
4.2.5 Objective 5:

Atlast, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

4.2.2 Proof of Concept

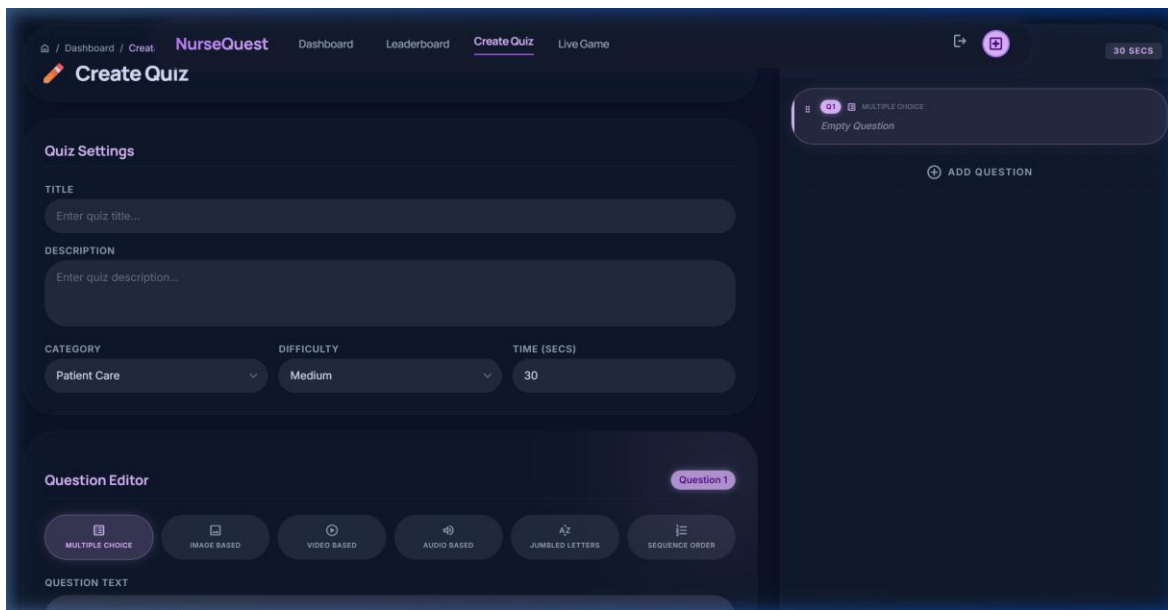
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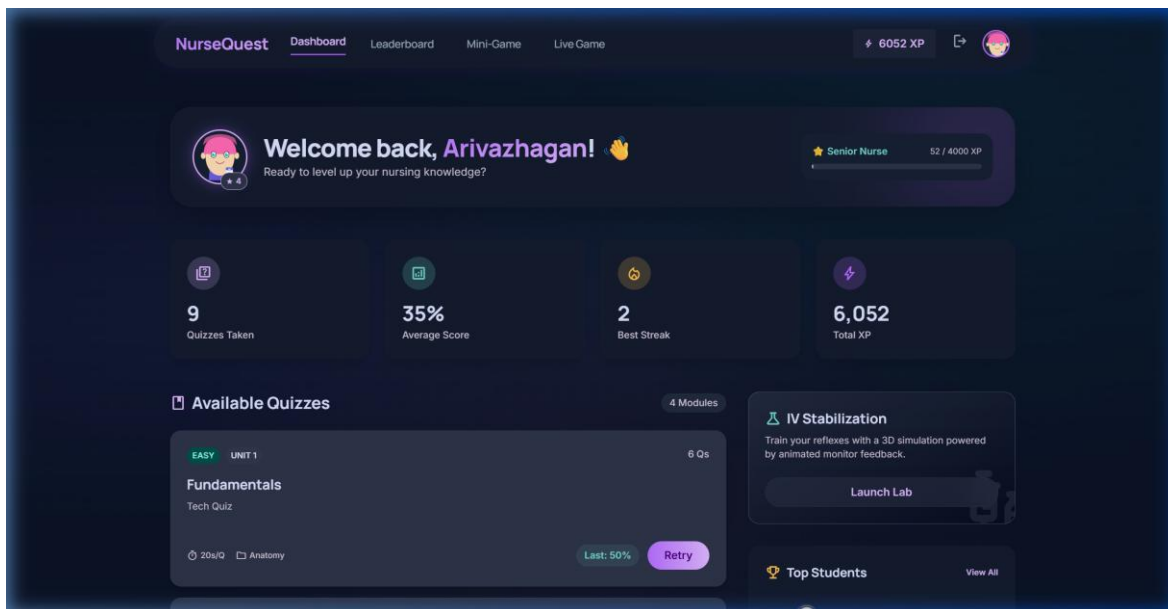
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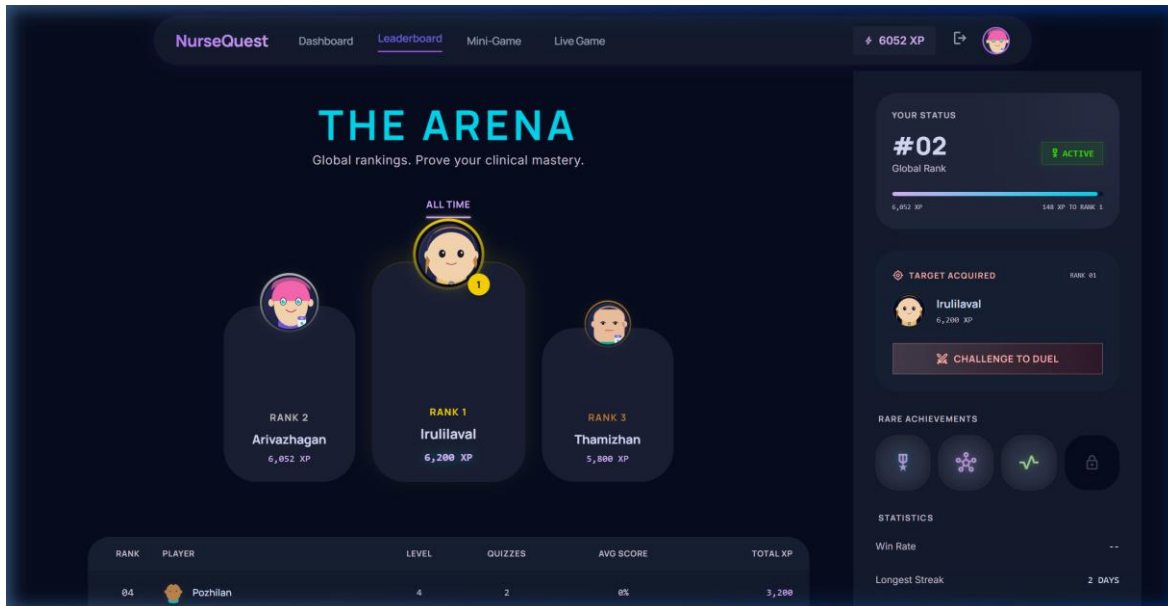
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Objective 4:

To implement live quiz multiplayer using Web socket with multiple device access and also showing live leaderboards with same question delivery improving student engagement and competitiveness



Objective 5:

Atlast, To implement and verify the production readiness of the app by combining security features like JWT, bcrypt, input validation and a scalable architecture for the real world deployment

```
> nursequest@1.0.0 dev
> concurrently -n BE,FE -c blue,magenta "npm run dev:backend" "npm run dev:frontend"

[BE] > nursequest@1.0.0 dev:backend
[BE] > cd backend && node server.js
[FE] > nursequest@1.0.0 dev:frontend
[FE] > cd frontend && npx vite --port 5173
[FE]
[BE] (node:35268) ExperimentalWarning: localStorage is not available because --localstorage-file was not provided.
[BE] (Use 'node --trace-warnings ...' to show where the warning was created)
[BE]
[BE] NurseQuest API Server
[BE] Running on http://localhost:3001
[BE] Socket.IO ready for real-time games
[BE]
[BE] Initializing Supabase database schema...
[BE] {
[BE]   severity_local: 'NOTICE',
[BE]   severity: 'NOTICE',
[BE]   code: '42P07',
[BE]   message: 'relation "users" already exists, skipping',
[BE]   file: 'parse_utilcmd.c',
```

```

BE] {
BE]   severity_local: 'NOTICE',
BE]   severity: 'NOTICE',
BE]   code: '42P07',
BE]   message: 'relation "idx_quiz_requests_status" already exists, skipping',
BE]   file: 'index.c',
BE]   line: '889',
BE]   routine: 'index_create'
BE] }
BE] ✓ Database schema initialized successfully
FE] VITE v8.0.10 ready in 513 ms
FE] → Local: http://localhost:5173/
FE] → Network: use --host to expose

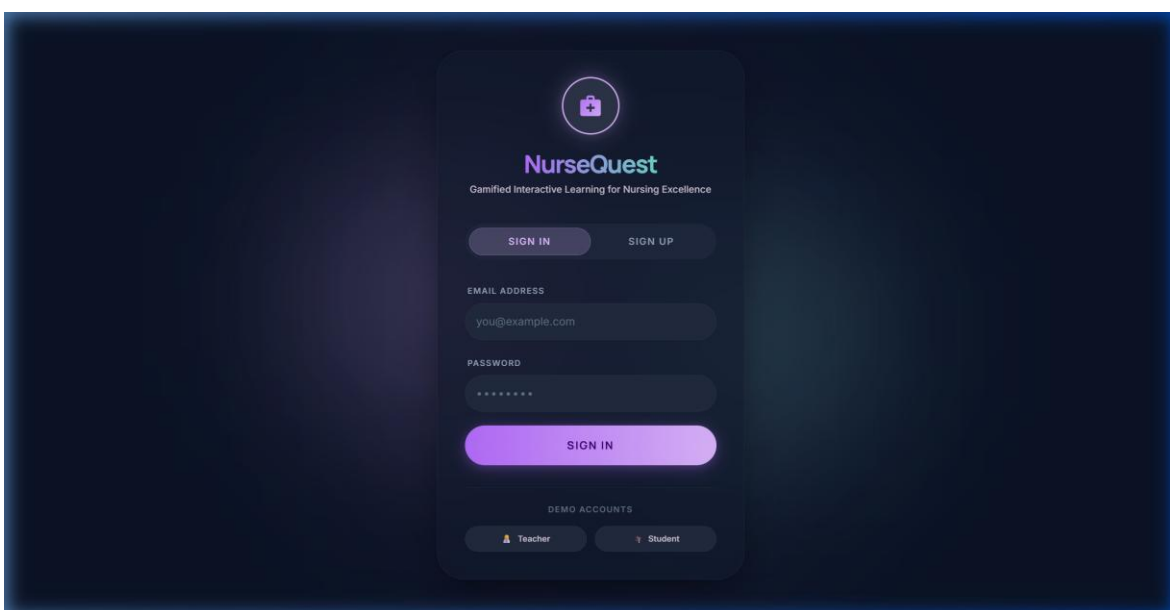
```

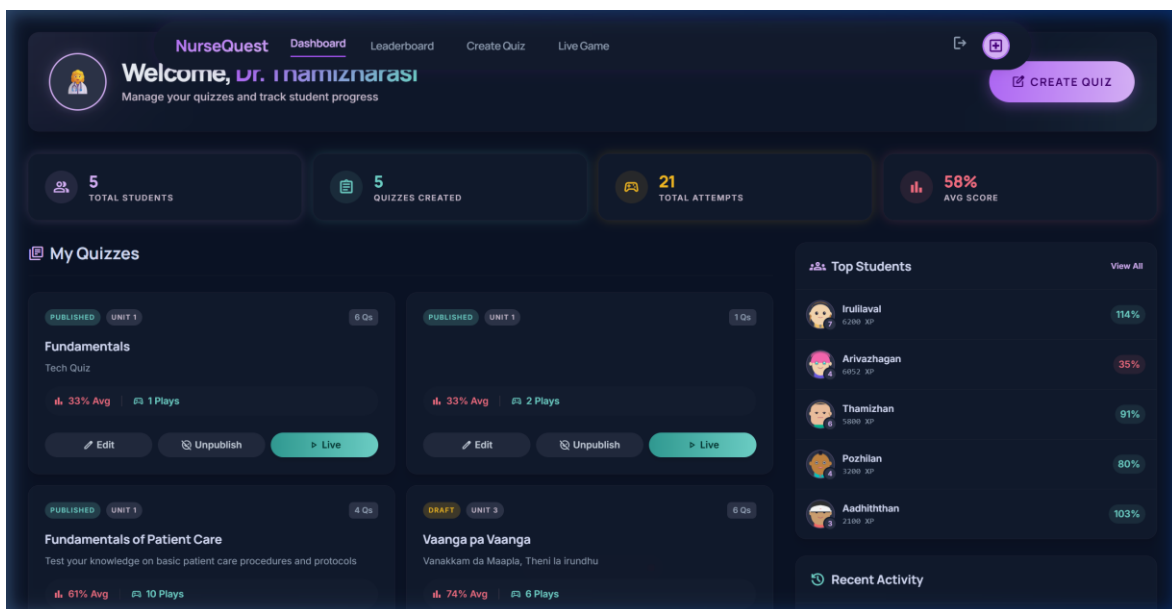
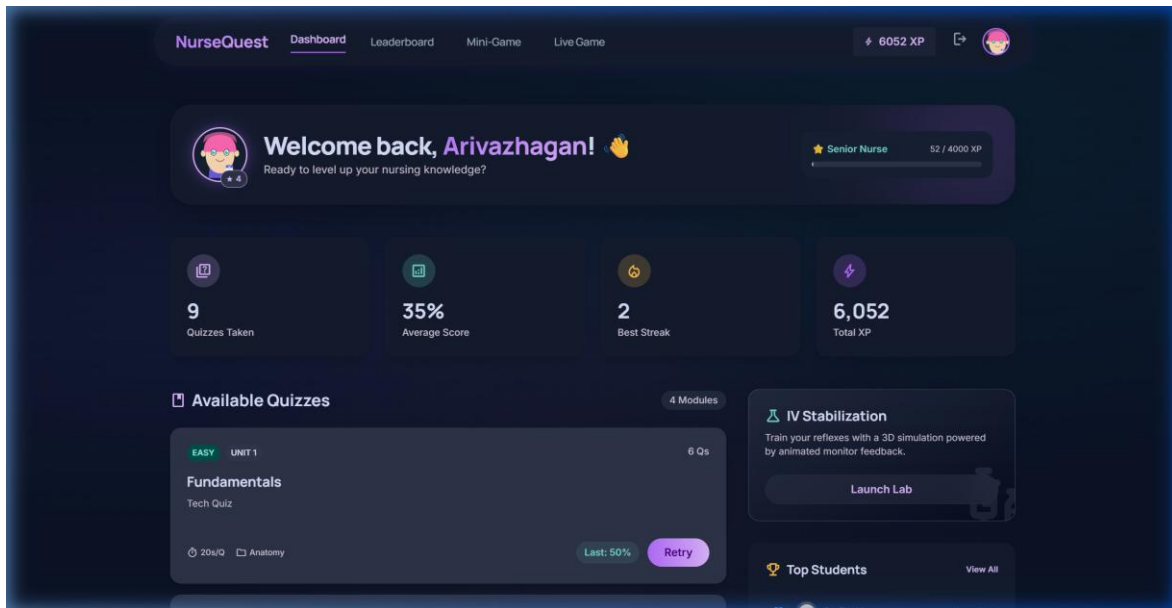
4.2.3 Formulation of Methods

The time bonus and live game scoring follows formulas for giving scores to the students. The formulas are derived from multiple quiz platforms and from now we have adapted to the Kahoot scoring system and we are giving the Formulas like that.

4.2.4 Prototype Design and Implementation

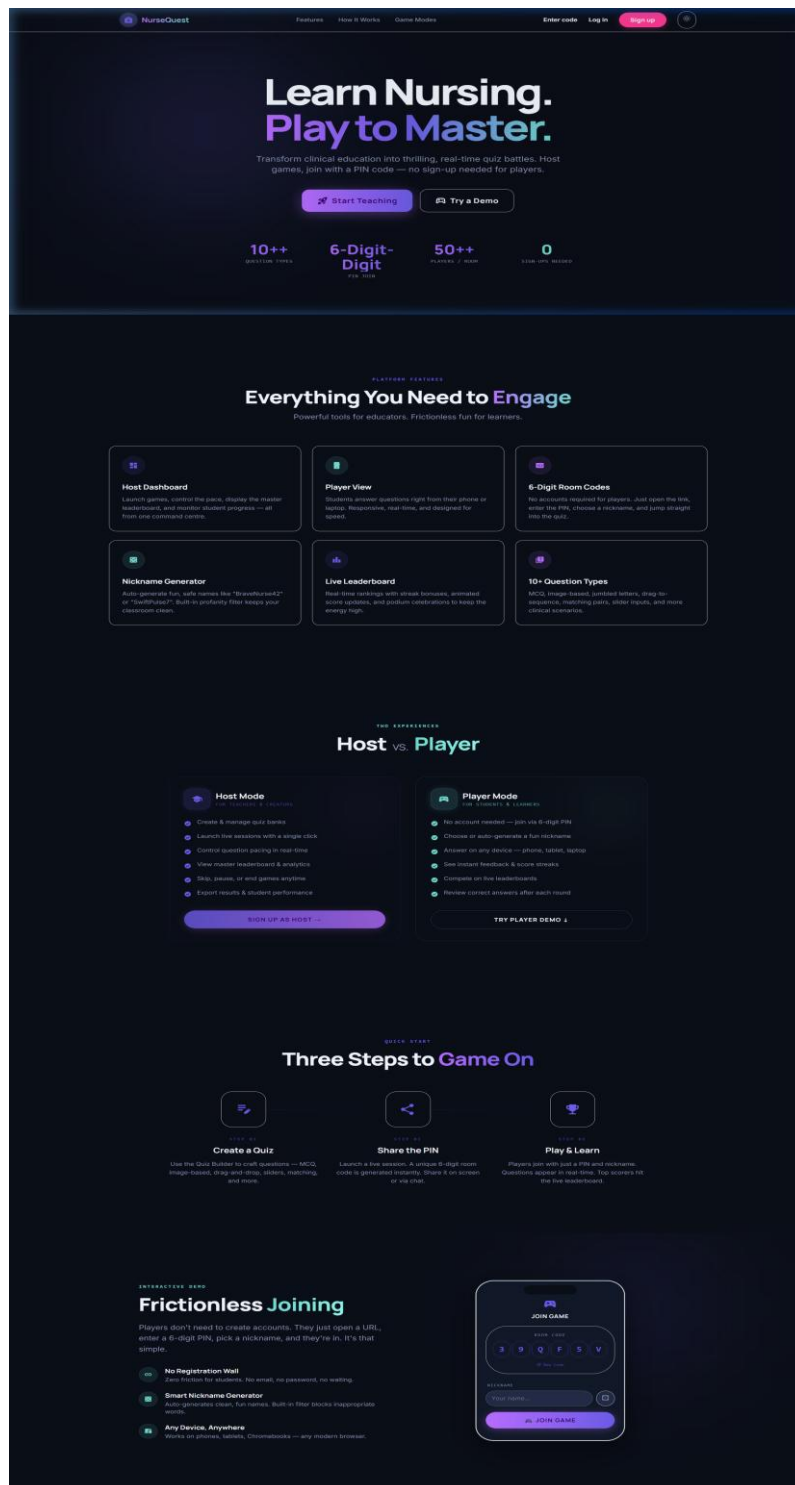
The First Implementative design is the Login page and the Role based Dashboards. We can see the images that is from the Login screen and Dashboards of the Different Roles



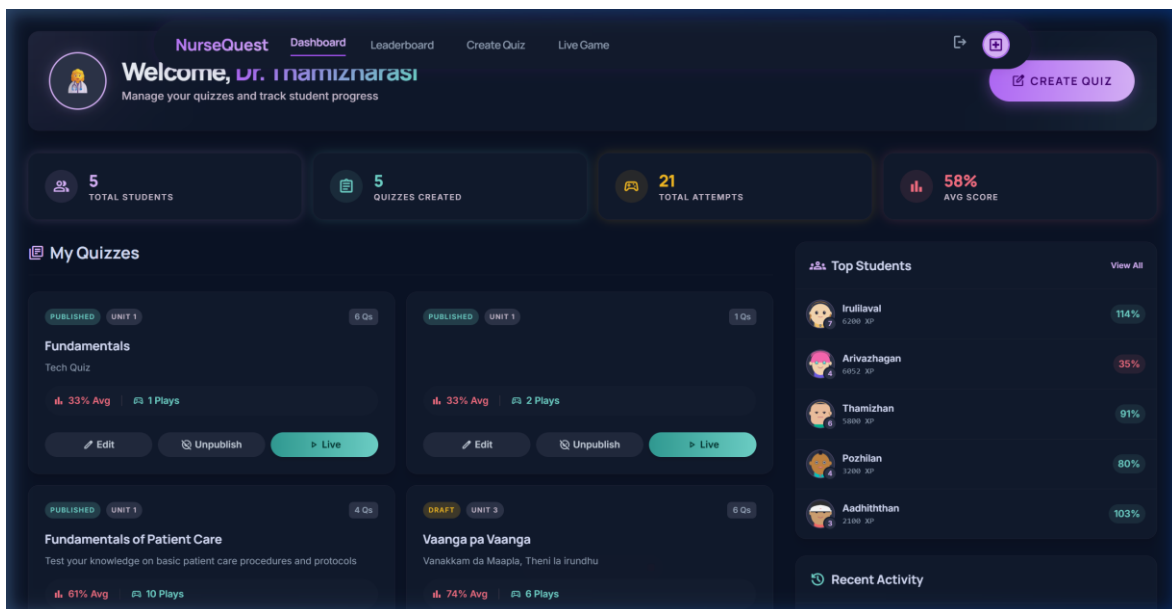
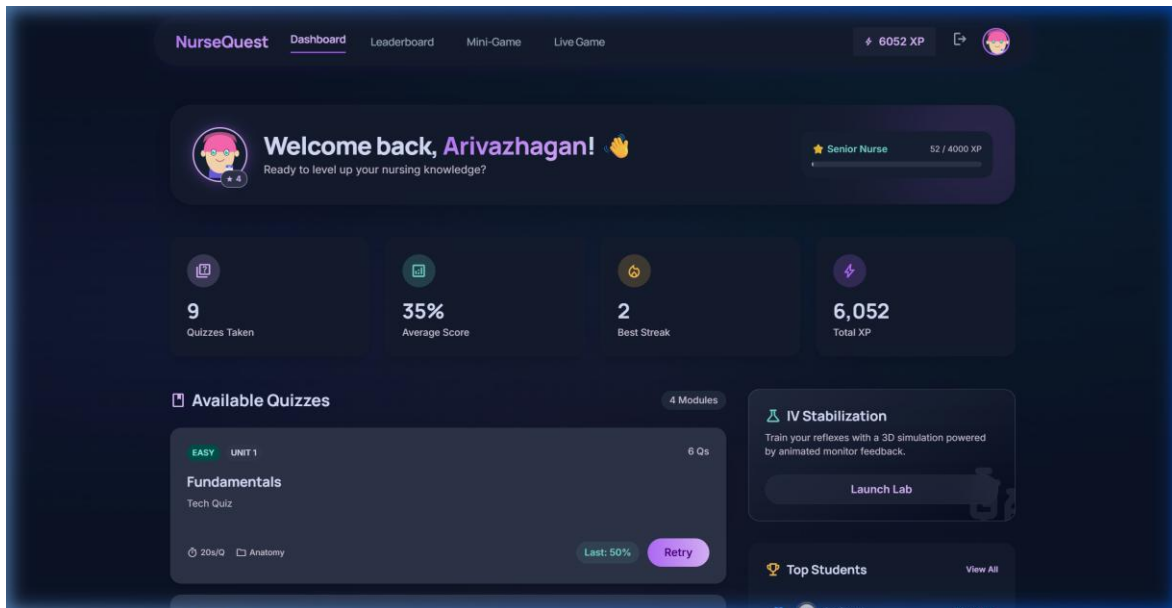


4.2.5 Working Model/ Dashboard/ Application (with screenshots)

The Landing page is designed as a Interactive attractive page which is scrollable and various elements are included in it.



The Dashboards are designed for comprehensive accessibility for the Users. The dashboards are divided for two roles, namely Teachers and Students



CHAPTER 5

RESULTS AND DISCUSSION

5.1 Data Collection

The Data Collection is done using the various quiz platforms and the data is obtained and Formulated for the implementation of scoring, and gamification elements and so on.

5.2 Computing Configuration

Windows 11

Intel Core I7

SSD 1 TB

165 Hz

5.3 Experimental Evaluation

5.3.1 Graph plots

In the stress tests, the Website performed well and the Graph plots are well scored. But the latency crashes when it touches thousands of users and this causes a delay in response.

5.3.2 Comparison of Results

The results have been taken from multiple students and about various questionnaire types. These results tell me that the Feedback of each questionnaire mode varies and this causes a friction among various questionnaire types.

5.3.3 Comparison with benchmark baseline.

The Benchmark baseline is nothing here. The app has not undergone any benchmark tests, but undergone stress tests. This makes the app not yet ready for Public deployment. For internal Phd Data Collection, It will be useful

CHAPTER 6

CONCLUSION

6.1 Research Findings and Conclusion

The development and deployment of the Website Nursequest gives us various findings among the students results and study behaviours. The results has shown that the results of the students improve to extents and daily engagement of students are there with the website. So the teachers also can draft and publish quizzes , so that after finishing every topic, the Teachers can crate a quiz and publish or host a live game.

6.2 Scope for Further Enhancement

Hosting in a paid service: If the Nursequest goes well, the Website will be used by thousands of users at a time. So that we are migrated to the a paid Hosting service for scalability

UI Redesigning: The UI redesigning is a process to enhance the Interactivity of the website among the students. So that, this process is a important process os that more users will use the Website

Generative AI: Large Language Models can be used for drafting Quizzes and implementing clinical scenarios and also for retriving feedbacks.

REFERENCES

Journal References: (a minimum of 15 works to be referenced and cited within text).

Web References:

Youtube

AI Platforms(Gemini,ChatGPT,Claude,etc.,.)

AI Builders (Google AI Studio, Stitch.io,etc.,.)

Online Forums (Pinterest,Reddit)

Online Quiz Platforms (Kahoot, Wayground, Mentimeter,etc.,.)

Publishers (ResearchGate, ebc.uk.com, etc.,.)

APPENDICES

APPENDIX-1:

Assign this quiz to a module for organized learning paths.

NurseQuest Dashboard Leaderboard Create Quiz Live Game

Question Editor Question 2

MULTIPLE CHOICE IMAGE BASED VIDEO BASED AUDIO BASED JUMBLED LETTERS SEQUENCE ORDER SLIDER MATCH PAIRS

QUESTION TEXT

Enter your question here...

SLIDER CONFIGURATION

Min Value: 0 Max Value: 100 Step: 1 Unit Label: e.g., mmHg, bpm

Correct Value

Enter the correct value

PREVIEW

0 100

No Module (Standard)

Assign this quiz to a module

NurseQuest Dashboard Leaderboard Create Quiz Live Game

Question Editor Question 1

MULTIPLE CHOICE IMAGE BASED VIDEO BASED AUDIO BASED JUMBLED LETTERS SEQUENCE ORDER SLIDER MATCH PAIRS

QUESTION TEXT

Enter your question here...

MATCH PAIRS

Students will see the right column shuffled. Define the correct pairs here.

Left Item 1	→	Right Item 1	×
Left Item 2	→	Right Item 2	×
Left Item 3	→	Right Item 3	×

APPENDIX –II: Only sample code/ Function call/ Abstract code of the project.

1. Scoring Algorithms:

```
function calculateScore(isCorrect, timeRemaining, totalTime, currentStreak, basePoints = 1000) {
```

```
  if (!isCorrect) {
```

```
    return {
```

```
      baseScore: 0,
```

```
      timeBonus: 0,
```

```
      streakBonus: 0,
```

```
      totalScore: 0,
```

```
      newStreak: 0,
```

```
      multiplier: 1
```

```
    };
```

```
  }
```

```
  const baseScore = basePoints;
```

```
  const timeFraction = Math.max(0, timeRemaining) / Math.max(1, totalTime);
```

```
  const timeBonus = Math.round(timeFraction * 500);
```

```
  const streakMultiplier = Math.min(currentStreak, 5);
```

```
  const streakBonus = streakMultiplier * 100;
```

```
  const totalScore = baseScore + timeBonus + streakBonus;
```

```
  return {
```

```
    baseScore,
```

```
    timeBonus,
```

```
    streakBonus,
```

```

    totalScore,
    newStreak: currentStreak + 1,
    multiplier: 1 + (streakMultiplier * 0.1)
  };
}

```

```

function calculateLiveScoreKahootStyle(isCorrect, elapsedMs, totalMs, maxPoints = 1000) {
  const safeTotalMs = Math.max(1000, Number(totalMs) || 0);
  const safeElapsedMs = Math.max(0, Math.min(Number(elapsedMs) || 0, safeTotalMs));
  const safeMaxPoints = Math.max(0, Number(maxPoints) || 0);

  if (!isCorrect) {
    return {
      maxPoints: safeMaxPoints,
      elapsedMs: safeElapsedMs,
      totalMs: safeTotalMs,
      timeFactor: 0,
      totalScore: 0
    };
  }

```

```

  const responseRatio = safeElapsedMs / safeTotalMs;
  const timeFactor = 1 - (0.5 * responseRatio);
  const totalScore = Math.max(0, Math.round(safeMaxPoints * timeFactor));

```

```

  return {
    maxPoints: safeMaxPoints,
    elapsedMs: safeElapsedMs,
    totalMs: safeTotalMs,
    timeFactor,

```

totalScore

};

}

WORKLOG

Attach the original signed copy of the Worklog, which is submitted at review review